

# Anu Ouseph

Working Student Software Developer

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## PROFILE

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Full-stack developer with with 2+ years of experience building real-time, interactive web applications. Strong focus on component architecture, WebSocket-driven interfaces, and visual/3D rendering using Three.js and GSAP. Comfortable working across the stack, Currently pursuing an M.Sc. in Research in Media Engineering.

## PROFESSIONAL EXPERIENCE

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**Product Engineer – Hoomans.co (India)**

06/2024 – 09/2025

- Built and maintained [Lyst.so](#), a real-time digital signage platform running on 500+ distributed displays and implemented WebSocket sync with reconnect logic and concurrent device state management (Vue.js, WebSockets, Go, PostgreSQL)
- Contributed frontend features to [TinkerHub.org](#), a tech community platform with 10K+ members, including TypeScript modules for user management, resource tracking, and event systems (Nuxt.js, TypeScript, GSAP)
- Built a custom interactive D3.js timeline visualization with scroll-driven animations and dynamic positioning for project timelines across multiple years (Vue.js, GSAP, D3.js) for [Hoomans.co](#)

## PROJECTS

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### [MIRa – Music Information Retrieval Analysis](#)

*Next.js · FastAPI · Python · Hugging face spaces · Vercel*

- Built a full-stack music analysis and comparison tool where users upload audio files and get back tempo, key, genre, mood, and instrument features in real time
- Designed a RESTful FastAPI backend deployed on Hugging Face Spaces (Docker) with a Next.js frontend
- Implemented music similarity comparison using weighted cosine similarity over MFCC, chroma, spectral centroid, and tempo feature vectors

### [Spatial – 3D Furniture Configurator](#)

*React.js · Three.js · react three fiber · Tailwind CSS*

- Browser-based 3D product configurator with real GLTF models, drag-to-rotate orbit controls, and real-time color and material switching
- Implemented scene graph traversal and material cloning for per-part customization on multi-mesh models
- Responsive UI with desktop side panel and mobile bottom sheet, and AR-ready architecture built to real-world scale

## Portfolio

*Vue.js · Three.js · GLSL*

- Interactive portfolio with custom GLSL shaders for mouse-driven image distortion and chromatic aberration effects
- Scroll-driven animations using GSAP ScrollTrigger with optimized rendering for smooth performance

## EDUCATION

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**M.Sc. Research in Media Engineering**

10/2025 – Present

*Technische Universität Ilmenau, Germany*

**B.Tech. Electronics Communication Engineering**

08/2018 – 07/2022

*Govt. College of Engineering, Kannur, India*

**Full-Stack Developer Program – Brototype**

11/2022 – 09/2023

*Intensive project-based training: JavaScript, Vue.js, React, Node.js, MongoDB, PostgreSQL*

## TECHNICAL SKILLS

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**Programming Languages:** JavaScript, TypeScript, Golang, Python

**Frontend:** React.js, Vue.js, Nuxt.js, Next.js, Three.js, GSAP, Tailwind CSS, Redux, Pinia

**Backend:** Node.js, Go, FastAPI, WebSockets, REST APIs, PostgreSQL, MongoDB, Supabase

**Tools:** Git, Docker, AWS (Lightsail, S3), Vercel, Hugging Face Spaces, CI/CD

**Languages:** English C1 (fluent) · German A2 (learning) · Malayalam (native)